

Gen AI Academy

APAC Edition

Build in APAC. Build for the world.



Participant Details

- a. Participant name: Sejal Kshirsagar
- b. Problem Statement: Build and deploy a production-ready AI agent using Google ADK and Gemini that performs a clearly defined task and is accessible via an HTTP endpoint.

Brief about the idea

Astronomy Explainer AI Agent 🌟

This project is an AI-powered astronomy assistant built using Google's Agent Development Kit (ADK) and Gemini.

The agent accepts user queries related to astronomy (e.g., black holes, planets, galaxies) and returns **simple, beginner-friendly explanations**.

It uses:

- Gemini for reasoning and explanation
- Wikipedia tool for factual grounding
- Cloud Run for scalable deployment

The goal is to make **complex space concepts easy to understand**.

Meeting the Build Criteria

This solution fully satisfies the project requirements:

- **✓ ADK-based Agent**
Implemented using Google ADK with modular agents
- **✓ Gemini Model Integration**
Uses Gemini (gemini-2.5-flash) for generating responses
- **✓ Single Defined Task**
→ Explaining astronomy concepts clearly
- **✓ Tool Usage**
→ Integrated Wikipedia tool for real-time knowledge retrieval
- **✓ HTTP Endpoint**
→ Built using FastAPI and deployed on Cloud Run
- **✓ Production Deployment**
→ Containerized using Docker and deployed serverlessly
- **✓ Scalable Architecture**
→ Cloud Run auto-scales based on demand

Opportunities

What makes this different?

- Focuses on **educational simplification**, not just answering
- Combines **tool-based retrieval + LLM reasoning**
- Uses **multi-agent workflow (research + explanation)**

Problem it solves:

- Astronomy is often **complex and hard to understand**
- Existing tools give **dense or technical answers**

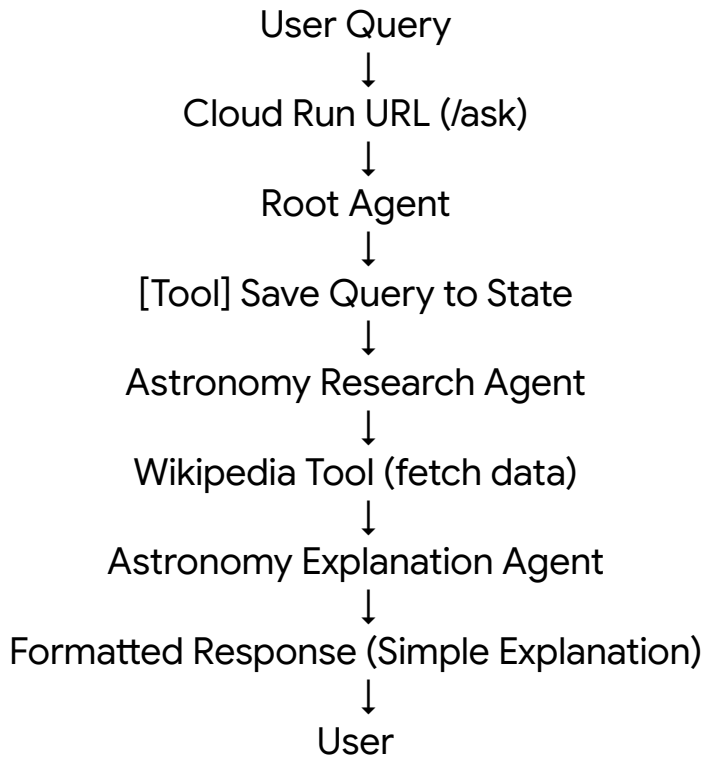
USP (Unique Selling Point):

- Converts **complex space topics** → **simple explanations**
- Uses **real-time knowledge (Wikipedia)**
- Modular agent design → easily extendable

List of features offered by the solution

-  Astronomy question answering
-  Beginner-friendly explanations
-  Wikipedia-powered research
-  Multi-agent pipeline (Research + Explain)
-  Fast responses using Gemini Flash
-  Cloud Run deployment (scalable & serverless)
-  HTTP API endpoint for easy integration

Process flow diagram or Use-case diagram



Technologies to be used in the solution

Programming Language: Python

AI Framework: Google ADK (Agent Development Kit)

LLM: Gemini 2.5 Flash

Tool Integration: LangChain + Wikipedia API

Cloud Platform: Google Cloud Run

Containerization: Docker

Authentication: IAM

Snapshots of the prototype

The screenshot displays the Agent Development Kit (ADK) interface. On the left, the 'astronomy_agent' is selected, and the 'Invocations' tab shows the query 'What are black holes?'. The main panel shows a session with ID 'c1330549-501d-496a-a2c8-9a0b4b163c26'. The session status is 'Token Streaming' (checked) and 'New Session' is available. The query 'What are black holes?' is entered in the input field. The session history shows the following actions:

- ⚡ add_query_to_state
- ✓ add_query_to_state
- ⚡ transfer_to_agent
- ✓ transfer_to_agent
- ⚡ wikipedia
- ✓ wikipedia

Snapshots of the prototype

The screenshot displays the Agent Development Kit (ADK) interface. On the left, a sidebar shows the 'astronomy_agent' selected under 'Invocations'. The main panel shows a chat session with the following content:

SESSION ID c1330549-501d-496a-a2c8-9a0b4b163c26 [Token Streaming] [New Session] [Trash] [Download]

⚡ wikipedia

✓ wikipedia

AI: A black hole is an astronomical body so compact that its gravity prevents anything, including light, from escaping. Albert Einstein's theory of general relativity predicts that any sufficiently compact mass will form a black hole. The boundary of no escape is called the event horizon.

Black holes typically form when very massive stars collapse at the end of their life cycle. After forming, they can grow by absorbing mass from their surroundings.

There are also supermassive black holes, which are the largest type, with masses hundreds of thousands to billions of times the mass of the Sun. Observational evidence indicates that almost every large galaxy has a supermassive black hole at its center, including our own Milky Way galaxy, which contains Sagittarius A*.

AI: Imagine a cosmic vacuum cleaner! A black hole is a region in space where gravity is so incredibly strong that nothing, not even light, can escape. This is because a huge amount of matter is packed into a tiny space.

The "point of no return" is called the **event horizon**. Once something crosses this boundary, it's trapped forever.

Black holes usually form when very massive stars collapse at the end of their lives. They can then grow by "eating" nearby gas and stars. The biggest ones, called **supermassive black holes**, live at the center of most galaxies, including our own Milky Way, where Sagittarius A* resides.

Type a Message...

[Attachment] [More] [Voice] [Video]



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Thank you

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